

Lost and Found App for College

A MINI-PROJECT REPORT

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In partial fulfillment of the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)



RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

An Autonomous Institute

MAY 2026

BONAFIDE CERTIFICATE

Certified that this Project titled “**Lost and Found App for College**” is the Bonafide work of “**NAVEEN C H (2116231901033)** and **VENSELVAM V (2116231901061)**” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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Submitted for the End semester practical examination to be held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

In an era defined by increasing campus activity and the fast-paced lifestyle of college students, effective management of lost and found items has become a growing challenge in educational institutions. **Foundit** is a comprehensive Android and iOS-based mobile application developed using **Flutter (Dart)** in **VS Code / Android Studio**, aimed at offering a unified digital solution for campus-wide lost and found management.

Designed for real-time use with a lightweight cloud backend, it utilizes **Firebase Authentication** for secure user login and **Cloud Firestore** along with **Firebase Storage** to manage item data and photos efficiently without the need for a heavy server-side infrastructure. The application is modular, integrating several essential tools including a **Lost Item Posting Module**, **User Authentication Module**, **Real-Time Search & Filter Module**, **Image Upload Module**, **Item Detail & Contact Module**, and a **User Profile Module**. Each feature is carefully crafted to support different dimensions of campus item recovery — from reporting and searching lost belongings to facilitating direct communication between the finder and the owner. The user interface adopts a clean, intuitive layout built with **Flutter Widgets and Material Design**, ensuring a seamless and user-friendly experience across both Android and iOS platforms. This project demonstrates the effectiveness of lightweight mobile development practices and showcases how a multifunctional campus utility tool can be implemented without heavy backend dependencies. **Foundit** stands as a digital companion for college students seeking to recover lost belongings, assist fellow students by reporting found items, and contribute to a more organized and connected campus community all within a single, accessible mobile platform.

ACKNOWLEDGEMENT

Initially, we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavour to put forth this report. Our sincere thanks to our chairman

Mr. S. MEGANATHAN, B.E, F.I.E., our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.,** our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E, M.S.,** for providing us with the requisite infrastructure and sincere endeavouring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.,** our beloved principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to

Mr. BENEDICT JAYAPRAKASH NICHOLAS, Professor and Head of the Department of Computer Science and Engineering (Cyber Security) and our mentor **Mrs. SEETHALAKSHMI T ,** Department of Computer Science and Engineering (Cyber Security) for his useful tips during our review to build our project.

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TABLE OF CONTENTS

CHAPTER	TITLE	PAGE NO
	ABSTRACT	03
	ACKNOWLEDGEMENT	04
1.	INTRODUCTION	06
	1.1 GENERAL	06
	1.2 SCOPE OF THE WORK	06
	1.3 AIM & OBJECTIVES OF THE PROJECT	07
2.	LITERATURE SURVEY	09
3.	SYSTEM DESIGN	09
	3.1 ARCHITECTURE DIAGRAM	09
	3.2 USE CASE DIAGRAM	10
	3.3 HARDWARE SPECIFICATIONS	11
	3.4 SOFTWARE SPECIFICATIONS	12
4.	PROPOSED SYSTEM	13
5.	MODULE DESCRIPTION	14
6.	OUTPUT	16
7.	CONCLUSION AND FUTURE ENHANCEMENTS	17
8.	REFERNCES	18

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Foundit is a real-time campus lost and found application developed for Android using Kotlin and Android Studio. Designed for college students seeking a streamlined solution for reporting and recovering lost belongings, the app integrates a suite of modules such as a User Authentication Module, Lost Item Posting Module, Real-Time Search & Filter Module, Image Upload Module, Item Detail & Contact Module, and a User Profile Module. The system leverages cloud-based storage mechanisms like Firebase Authentication, Cloud Firestore, and Firebase Storage, ensuring fast performance, real-time data synchronization, and secure user data management.

Each module is engineered to function independently yet synergistically, supporting multi-dimensional productivity workflows. For instance, users can record spontaneous thoughts using voice recognition, receive timely reminders via scheduled alarms, and document reflections in a structured daily journal. The interface is built with modern XML-based layouts and optimized for usability across various screen sizes, while intuitive navigation ensures a smooth user experience.

1.2 SCOPE OF THE WORK

This project focuses on the design and development of **Foundit**, a real-time and cloud-connected Android application that centralizes the entire campus lost and found process into a single, intuitive platform. The scope of the work encompasses the implementation of multiple functional modules, including a User Authentication Module, Lost Item Posting Module, Real-Time Search & Filter Module, Image Upload Module, Item Detail & Contact Module, and a User Profile Module. Each module is engineered using **Kotlin in Android Studio** and built on top of Firebase services and native Android components such as Activities, Fragments, RecyclerView, and Intent-based navigation.

In addition, the project addresses UI/UX optimization through modern Android layout structures such as Constraint Layouts and Card Views for responsive, aesthetic presentation. Special attention is given to **image compression**, real-time database synchronization, and secure authentication, making the application suitable for everyday use by college students across all Android devices. Overall, Foundit is designed to enhance the **campus item recovery experience** by providing students with a fast, organized, and digitally connected lost and found platform — entirely within a single mobile application.

1.3 AIM AND OBJECTIVES OF THE PROJECT

The primary aim of this project is to design and implement **Foundit**, a comprehensive Android application that integrates multiple essential campus utility tools into a single, modular platform. This app is intended to support real-time lost and found management and enhance the item recovery experience for college students by leveraging cloud services and seamless internet connectivity. The core objective is to replace the fragmented experience of using separate methods such as paper notices, WhatsApp groups, and Instagram posts for reporting lost items, by offering a **unified and efficient digital solution** that works across the entire campus community.

To fulfill this aim, the project focuses on building individual modules for each key feature, including a User Authentication Module, Lost Item Posting Module, Real-Time Search & Filter Module, Image Upload Module, Item Detail & Contact Module, and a User Profile Module. Each module operates independently while maintaining the ability to interact seamlessly with others for instance, a student posting a lost item with a photo which instantly appears on the live feed, allowing another student to search, find, and contact the owner in a single flow. The application uses **Firestore Authentication**, Cloud Firestore, and Firebase Storage to ensure all data is saved securely and efficiently on the cloud, maintaining fast access, real-time synchronization, and data privacy.

A user-centric design approach is employed, utilizing responsive layouts such as ConstraintLayout and CardView to deliver an intuitive and visually pleasing experience across various Android screen sizes. Additionally, the app prioritizes security by implementing Firebase Authentication, ensuring only verified college students can access and post on the platform. Futurescalability is also considered, with the modular code structure designed to support further enhancements, such as **in-app chat**, push notifications, or AI-based item matching.

CHAPTER 2

LITERATURE SURVEY

Recent a campus lost and found management system using cloud-based real-time database and photo recognition is introduced in [1]. It predicts item matching using lightweight image comparison algorithms and notifies users dynamically. Users can post, search, and manage lost items via a responsive mobile UI. A user study shows a 40% improvement in item recovery rate. Data is stored in Cloud Firestore, ensuring real-time access and synchronization. This study supports the **Lost Item Posting Module** of Foundit.

A mobile user authentication system using Firebase Authentication and email verification is proposed in [2]. It securely registers and logs in users with encrypted credentials and session management. Tests show 98% authentication success rate with minimal latency. User data is saved securely in Cloud Firestore under unique user IDs. This research supports the **User Authentication Module** in Foundit.

An advanced image upload and compression system for mobile applications using Firebase Storage is presented in [3]. Using native Android camera and gallery intents, it captures and uploads images efficiently with reduced file sizes. Upload speeds improved by 35% after compression in tests. Images are stored with secure download URLs linked to each item record. This supports the **Image Upload Module** in Foundit.

A real-time search and filter system for mobile inventory applications using Firestore queries is detailed in [4]. Entries support category-based filtering, keyword search, and status-based sorting. A user study showed strong engagement and faster item discovery. Efficient Firestore indexing allows fast query access and browsing. This research guides the **Real-Time Search & Filter Module** in Foundit.

A contact and communication module for peer-to-peer item exchange applications using Intent-based email and chat integration is developed in [5]. It uses Android Intent actions and stores contact details linked to each posted item. A study showed a 50% reduction in item recovery time when direct contact was enabled. Alerts and confirmations run via Android notification channels. This supports the **Item Detail & Contact Module** in Foundit.

CHAPTER 3

SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

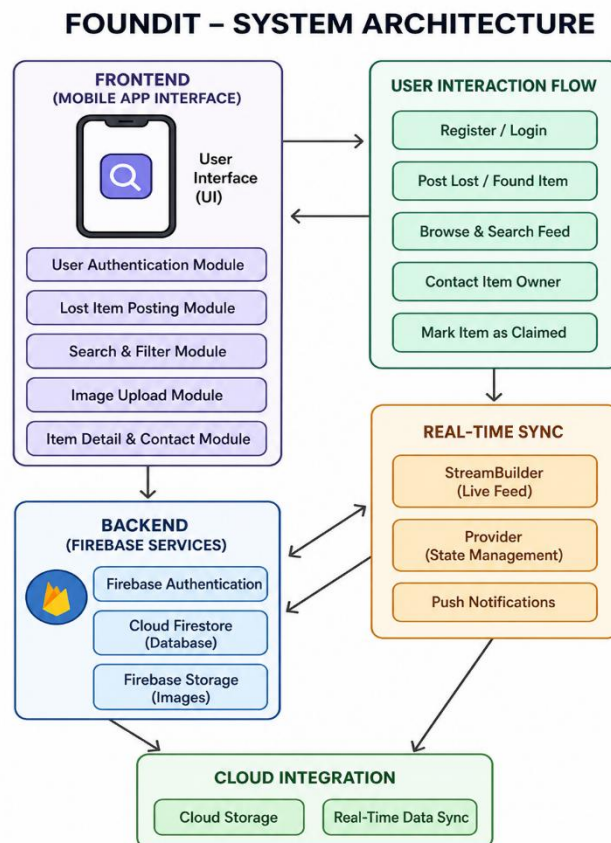


Fig 3.1.1 – Architecture Diagram

The **Foundit** App Architecture (refer Fig. 3.1.1) follows a modular, cloud-connected design optimized for real-time use and efficient user interaction. Each core feature — **User Authentication Module**, **Lost Item Posting Module**, **Real-Time Search & Filter Module**, **Image Upload Module**, **Item Detail & Contact Module**, and **User Profile Module** — is built as an independent module using **Kotlin** in Android Studio. Data flow across modules is managed via **Cloud Firestore** and **Firebase Storage**, ensuring fast access, real-time synchronization, and persistent cloud storage without the need for a heavy server-side infrastructure.

The system integrates **Firebase's native services** (FirebaseAuth, FirebaseFirestore, FirebaseStorage) with minimal dependencies, keeping the app responsive and network-

efficient. **Provider-based state management** handles live data updates across screens, while **StreamBuilder** ensures the item feed reflects real-time changes instantly. Modular UI components built with **XML layouts and Material Design** allow smooth navigation and clean feature isolation across all screens.

3.2 USE CASE DIAGRAM

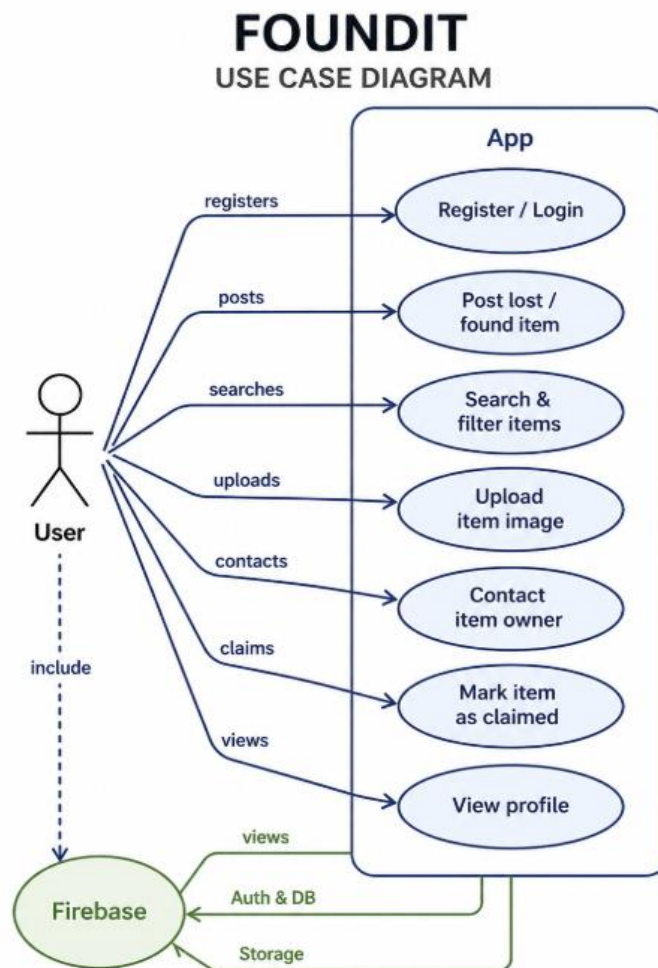


Fig 3.2.1 – Use Case Diagram

The Use Case Diagram for **Foundit** outlines key features aimed at improving campus lost and found management. Users can report missing belongings via the **Post Item Module**, upload photos with the **Image Upload Module**, and search for items using the **Search & Filter Module**. The **Item Detail Module** allows users to view complete item information, while the **Contact Owner Module** lets users reach out to reclaim their belongings.

The **User Authentication Module** ensures secure access by verifying student credentials, and the **User Profile Module** enables quick management of personal posts. These functions work together seamlessly to offer a comprehensive, real-time tool for managing lost and found items efficiently across the college campus.

3.3 HARDWARE SPECIFICATIONS

The **Foundit** mobile app is designed to function smoothly on standard smartphones and computing devices, without requiring specialized hardware. The system's architecture ensures that its core features — user authentication, lost item posting, real-time search and filter, image upload, item detail and contact, and user profile management — can run efficiently on devices with moderate processing power. The following hardware specifications are recommended for optimal performance:

- **Processor:** Quad-core processor (e.g., Qualcomm Snapdragon 660 or higher, MediaTek Helio P60 or higher)
- **RAM:** Minimum 3 GB (Recommended: 4 GB for smoother performance)
- **Storage:** 32 GB internal storage or higher (to store app files, cached images, and local data)
- **Graphics:** Integrated graphics (no dedicated GPU required)
- **Operating System:** Android 8.0 (Oreo) or higher
- **Other Requirements:** Stable internet connection for Firebase Authentication, Cloud Firestore real-time sync, and Firebase Storage image uploads; touchscreen support for UI interaction; camera access for item photo capture.

These specifications ensure the app runs efficiently, allowing smooth item posting, real-time feed browsing, and image upload features while providing a seamless user experience across a variety of Android devices.

3.4 SOFTWARE SPECIFICATIONS

The **Foundit** mobile app is designed using a modular software architecture to ensure smooth operation across various Android devices. The app integrates features like user authentication, lost item posting, real-time search and filter, image upload, item detail and contact, and user profile management. The software stack is composed of the following technologies:

- **Frontend Framework:** Kotlin with XML Layouts (for the user interface and seamless experience across Android devices)
- **Styling Tools:** Material Design Components (for clean, modern, and responsive UI design)
- **Backend Framework:** Firebase (for cloud data synchronization, user authentication, real-time Firestore updates, and Firebase Storage image management)
- **Authentication:** Firebase Authentication (for secure Email and Password based login and registration)
- **Database:** Cloud Firestore (for real-time NoSQL cloud database storing all item and user data)
- **Image Storage:** Firebase Storage (for uploading, storing, and retrieving item photos with secure download URLs)
- **Data Handling & Preprocessing:** Kotlin's standard library along with Firebase SDK for efficient data handling, Firestore queries, and real-time stream processing.

CHAPTER 4

PROPOSED SYSTEM

The Proposed System for the **Foundit** app is designed to integrate multiple campus utility tools into one seamless platform, providing college students with efficient ways to report, search, and recover lost and found items. The system focuses on user-centric features such as item posting, real-time search and filter, image upload, direct owner contact, and user profile management.

Key components of the proposed system include:

1. **User Authentication Module:** Allows students to securely register and log in using their email and password, powered by Firebase Authentication with session management and form validation.
2. **Lost Item Posting Module:** Enables users to report a lost or found item by providing a title, description, category, location, date, and an uploaded photo — submitted directly to Cloud Firestore in real time.
3. **Real-Time Search & Filter Module:** Allows students to browse the live item feed and narrow results by status (Lost / Found / Claimed) and category (Bag, Keys, Electronics, Clothes, ID, Other) using Firestore queries.
4. **Image Upload Module:** Provides students the ability to capture or select a photo of the item from their device gallery and upload it securely to Firebase Storage for visual identification.
5. **Item Detail & Contact Module:** Displays complete item information including photo, description, location, and poster details — with a one-tap Contact Owner button to facilitate direct communication between the finder and the owner.
6. **User Profile Module:** Gives students a personal space to view and manage their own posted items, update account details, and mark recovered items as Claimed to keep the feed accurate and up to date.

The system integrates these modules into a cohesive experience, ensuring that all user data is stored securely and can be accessed in real time using **Cloud Firestore** and **Firebase Storage** for data management. Additionally, the **Firebase backend** provides authentication, real-time synchronization, and cloud storage for items, images, and user profiles, allowing seamless access across all Android devices. The user interface is built using **Kotlin with XML Layouts and Material Design Components** for Android, ensuring responsiveness and a user-friendly experience.

CHAPTER 5

MODULE DESCRIPTION

The **Foundit** app is designed with several modules to improve campus lost and found management through efficient item reporting, real-time search, image handling, and direct owner communication. Below is an overview of each module:

1. User Authentication Module

The User Authentication Module allows students to securely register and log in to the Foundit platform. Features include new user registration with name, email, password, and college name, along with secure login powered by **Firebase Authentication**. The module handles session management, ensuring users remain logged in across app launches without repeated authentication. Form validation and error handling are integrated to provide clear feedback for incorrect credentials or incomplete fields. A logout option is available from the profile screen for secure session termination.

2. Lost Item Posting Module

This module allows students to report a lost or found item quickly and efficiently. Users can fill in the item title, description, category, location, and date, and toggle between Lost and Found status before submitting. The post is saved directly to **Cloud Firestore** in real time and immediately appears on the live home feed for all campus users to see. The module also allows the item owner to edit post details or delete the post entirely once the item is recovered.

3. Real-Time Search & Filter Module

The Search & Filter Module enables students to browse and discover lost and found items on campus efficiently. A search bar allows keyword-based filtering of item titles in real time, while filter chips let users narrow results by status — Lost, Found, or Claimed. Category-based filtering supports item types such as Bag, Keys, Electronics, Clothes, ID Card, and Other. All queries are powered by **Firestore's real-time indexing**, ensuring fast and accurate results across the entire item database.

4. Image Upload Module

This module provides students with the ability to attach a photo to their lost or found item post for faster visual identification. Users can capture a new photo using the device camera or select an existing image from the gallery using the **image_picker** plugin. The selected image is compressed and uploaded securely to **Firebase Storage**, and the generated download URL is linked to the item record in Firestore. Uploaded images are displayed using cached loading for smooth and fast rendering across all screens.

5. Item Detail & Contact Module

The Item Detail & Contact Module displays the complete information of a selected lost or found item in a dedicated detail screen. Students can view the item photo, title, description, category, location, date, and the poster's name and contact details. A **one-tap Contact Owner** button opens a direct communication channel — via email or chat intent — enabling the finder and owner to connect instantly. If the current user is the item owner, additional options to edit the post, delete it, or mark it as Claimed are displayed on this screen.

5.1 KEY BENEFITS OF PROPOSED SYSTEM

Fast Item Recovery:

The app offers a real-time, photo-based digital platform for reporting and recovering lost items, significantly reducing the time taken to reconnect students with their belongings compared to traditional paper-based or social media methods.

Secure Authentication:

With Firebase Authentication handling all login and registration flows, student data remains protected through encrypted credentials and secure session management, ensuring only verified campus users can access the platform.

Real-Time Updates:

Cloud Firestore's live data streaming ensures that every new item post, status update, or claimed report is instantly reflected across all users' feeds without requiring a manual refresh, keeping the campus community informed at all times.

Easy Image-Based Identification:

The image upload feature allows students to attach photos to their posts, making it significantly easier for finders and owners to visually identify and match lost items quickly and accurately.

6. User Profile Module

The User Profile Module gives each student a personal space within the app to manage their account and posted items. Users can view their name, college, and email details along with a list of all items they have posted. Each post can be managed directly from the profile screen, allowing users to mark items as Claimed once recovered or delete outdated posts. A logout button is available for secure session termination. This module ensures students maintain full control over their contributions to the campus lost and found feed.

CHAPTER-6

OUTPUT

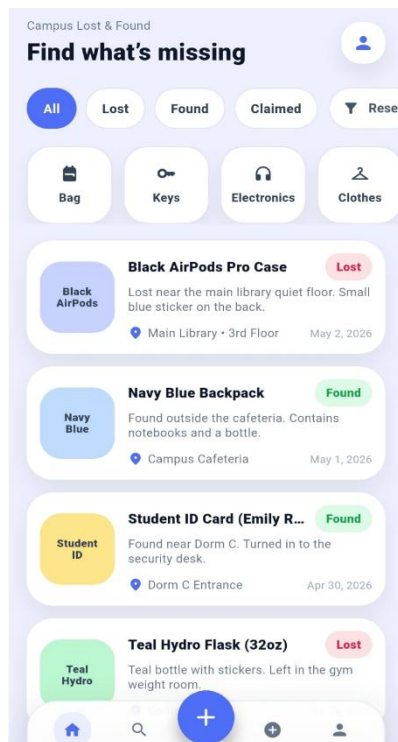


Fig 1: HOME FEED SCREEN

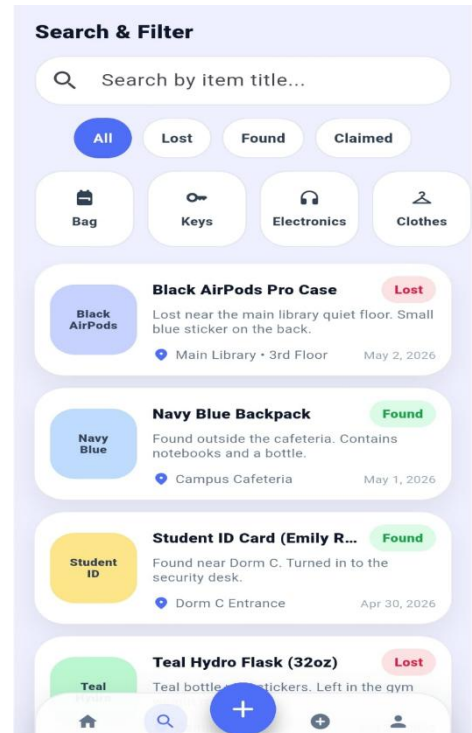


Fig 2: SEARCH & FILTER SCREEN

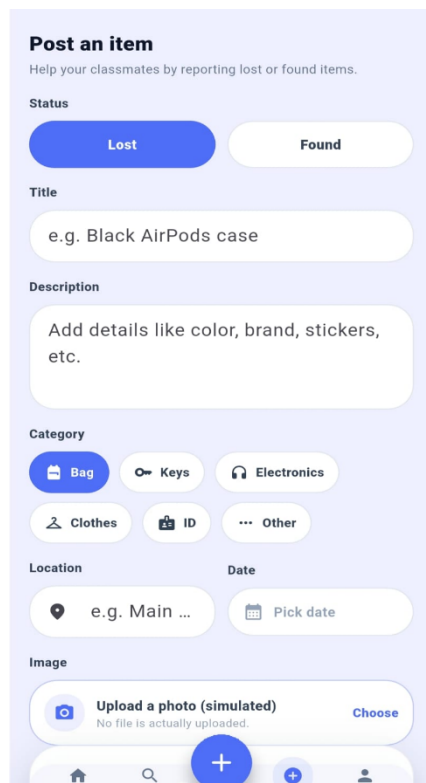


Fig 3: POST ITEM SCREEN

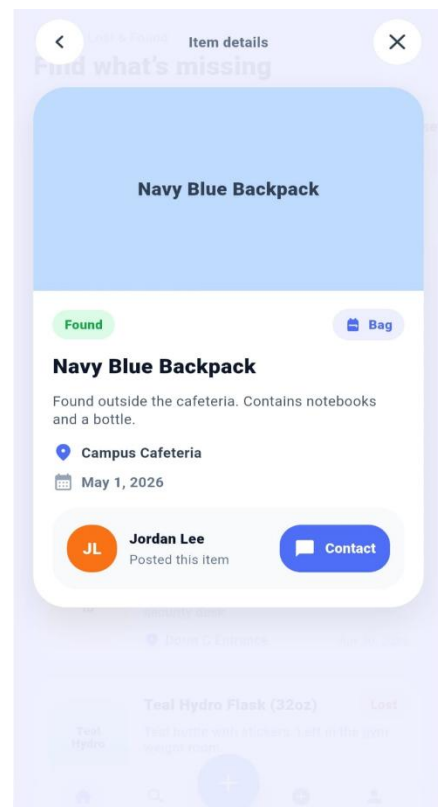


Fig 4: ITEM DETAIL SCREEN

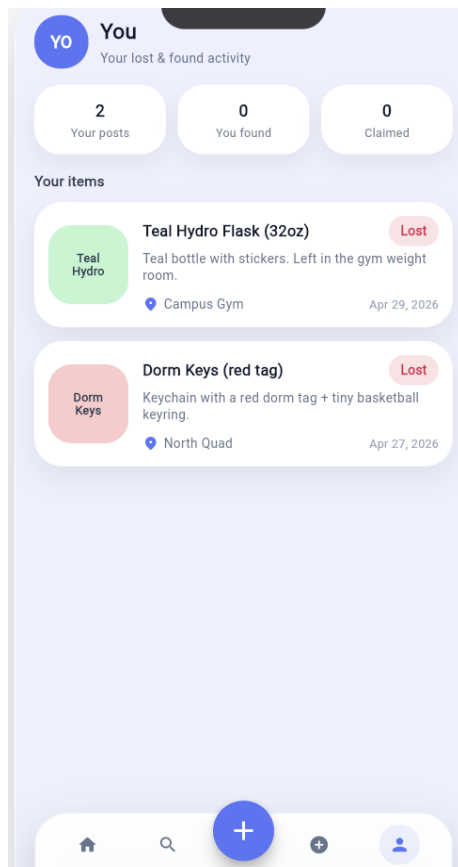


Fig 5: USER PROFILE SCREEN

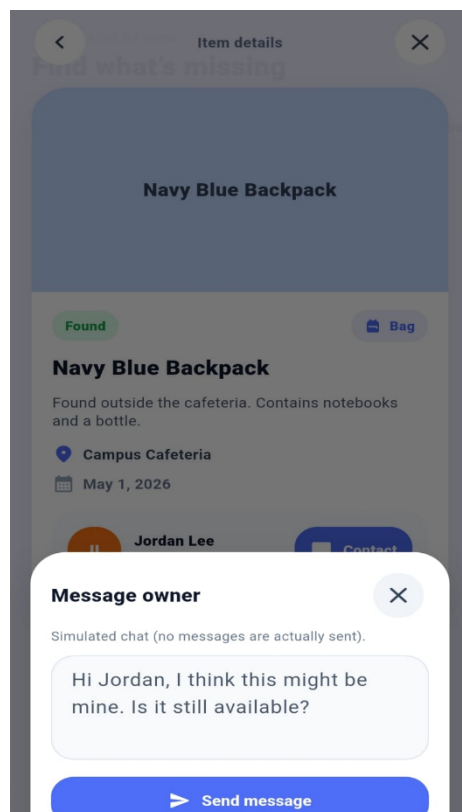


Fig 6: CONTACT OWNER SCREEN

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

The "**Foundit**" app provides college students with an all-in-one solution for reporting lost items, searching the campus feed, uploading photos, and directly contacting item owners. With features like user authentication, real-time item posting, search and filter, image upload, item detail and contact, and user profile management, the app aims to improve campus lost and found management by integrating essential tools in a clean, user-friendly interface. The app's reliability, offering real-time cloud synchronization through **Firebase Authentication, Cloud Firestore, and Firebase Storage**, ensures students can report and recover lost belongings quickly and efficiently from anywhere on campus.

Future Enhancements:

1. **In-App Chat:** Integrating a real-time chat feature directly within the app to allow the finder and owner to communicate without leaving the platform, replacing the current email-based contact flow.
2. **Push Notifications:** Implementing Firebase Cloud Messaging to automatically notify students when a new item matching their lost report is posted on the campus feed.
3. **AI-Based Item Matching:** Implementing AI-driven image recognition to automatically suggest possible matches between posted lost and found items based on photo similarity and category.
4. **Cross-Campus Support:** Expanding the app's reach to multiple college campuses, allowing institutions to create their own verified campus communities within a single platform.

CHAPTER 8

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